

COSIMA Cookbook: a computational learning module for analysing ocean and sea-ice model output

Navid C. Constantinou^{1,12*}, Julia Neme^{2*}, Wilton Aguiar², Matthis Auger¹⁴, Ashley J. Barnes^{4,8}, Romain Beucher³, Dhruv Bhagtani⁹, Christopher Yit Sen Bull⁴, Hannah Dawson¹⁴, Noah Day¹², Fabio Boeira Dias¹¹, Edward Doddridge¹⁴, Anupiya Ellepola², Matthew England¹¹, Denisse Fierro-Arcos¹⁴, Angus Gibson², Aidan Heerdegen³, Andy McC. Hogg^{3,4}, Ryan M. Holmes⁵, Wilma Huneke², Jemma Jeffree^{2,4}, Andrew E. Kiss², Minghang Li³, Josué Martínez-Moreno¹⁰, Jan Jaap Meijer¹⁴, Thomas Moore⁷, Ruth Moorman⁶, Adele Morrison², James Munroe¹, Aditya Narayanan¹³, Micael Oliveira³, Ellie Q. Y. Ong^{4,8}, Max Proft³, Madelaine G. Rosevear^{4,14}, Christina Schmidt¹⁰, Taimoor Sohail¹², Paul Spence^{4,14}, Dougal T. Squire³, Anton Steketeer³, Charles Turner³, Felipe Vilela da Silva¹⁴, Marc White³, Luwei Yang², Claire Yung², and Jan Zika¹¹

¹ 2i2c ² Australian National University, Australia ³ Australian National University, Australia's Climate Simulator (ACCESS-NRI), Australia ⁴ Australian Research Council Centre of Excellence for the Weather of the 21st Century, Australia ⁵ Bureau of Meteorology, Australia ⁶ California Institute of Technology, USA ⁷ Commonwealth Scientific and Industrial Research Organisation, Australia ⁸ Monash University, Australia ⁹ Princeton University, USA ¹⁰ British Antarctic Survey, UK ¹¹ University of New South Wales Sydney, Australia ¹² University of Melbourne, Australia ¹³ University of Southampton, UK ¹⁴ University of Tasmania, Australia * These authors contributed equally.

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Software

- [Review](#) 
- [Repository](#) 
- [Archive](#) 

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Summary

The COSIMA Cookbook is an open computational learning module for analysing ocean and sea-ice model output in Jupyter notebooks. It has been developed by the Consortium for Ocean-Sea Ice Modelling in Australia (COSIMA; <https://cosima.org.au>) as a community resource for researchers, students, and practitioners working with large gridded datasets, especially output from the ACCESS ocean-sea ice model configuration ([Kiss et al., 2020](#)). The repository combines introductory tutorials with worked analysis examples, all exposed through a browsable Sphinx documentation site ([Figure 1](#)) and backed by a citable archived release ([COSIMA contributors, 2026](#)).

COSIMA
Cookbook

Go

Navigation

Lessons

Cooking Lessons 101

- Basics
- Advanced

Recipes

Appetisers

Mains

Local Dishes

Resources

Contributing

- Getting started with git
- Working on your recipe
- Tips for making a good recipe
- Submitting a Pull Request
- Reviewing existing Pull Requests

Notebook Guidelines

- Thumbnails
- Images in notebooks

GitHub Repository

Appetisers (easy)

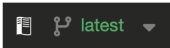
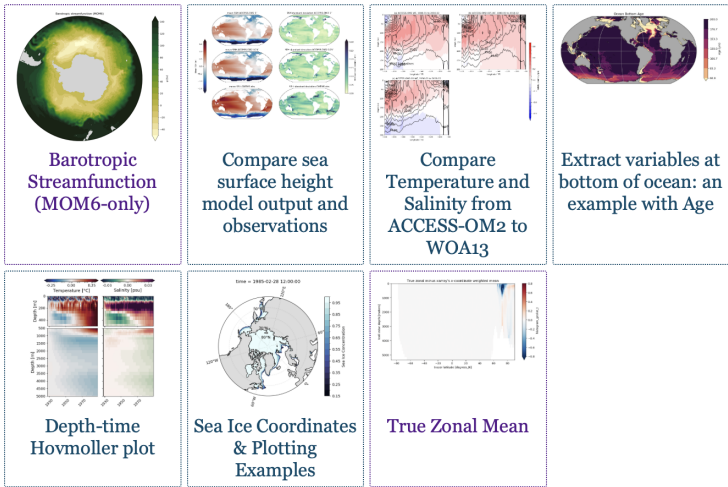


Figure 1: The COSIMA Cookbook documentation website, showing the browsable Sphinx-based landing page used to navigate tutorials and recipes. The live site is available at <https://cosima-recipes.readthedocs.io>.

The website sections are deliberately named using a gastronomy theme, consistent with the “Cookbook” concept. “Cooking Lessons 101” refers to tutorials that teach generic, transferable techniques; “Appetisers” are easier recipes that provide a good entry point after the tutorials; “Mains” are more elaborate and advanced analysis examples; and “Local Dishes” collect recipes for regional model configurations (Barnes et al., 2024). Together these sections present the documentation as a browsable collection of lessons and recipes gathered into a single cookbook.

Pedagogy and instructional structure are central to the project. “Cooking Lessons 101” introduces generic skills such as loading model output, working with labelled arrays, plotting, and interacting with shared data catalogues. These tutorials lead into domain-focused “recipes”: self-contained notebooks showing how to perform concrete diagnostics and analyses on ocean and sea-ice datasets. The current repository contains dozens of notebooks grouped into introductory, advanced, and region-specific collections, giving learners an incremental path from first contact with the data ecosystem to adaptation of full workflows for their own science questions.

Within this structure, the tutorials include examples of loading model output through intake-based catalogues (Intake developers, 2026) and incorporating Cartopy (Met Office, 2010–2015) for map-based visualisation. The recipe collection also includes a Lagrangian particle-tracking workflow built with Parcels (Lange & Seville, 2017), as well as analyses that use xgcm (xgcm Developers, 2025) to handle variables defined on staggered finite-volume grids.

53 The Cookbook grew from a practical need inside the COSIMA community: the tools used
54 to analyse modern ocean model output are powerful, but the gap between package-level
55 documentation and a reproducible end-to-end workflow remains large. Users need to
56 understand not only Python and Jupyter (Kluyver et al., 2016), but also how to navigate
57 high-dimensional model output, shared high-performance computing environments, and
58 domain-specific analysis conventions. The Cookbook addresses that gap by packaging
59 reusable workflows in the same medium in which users actually work.

60 Statement of Need

61 Ocean and climate model analysis has a steep entry cost. New users must learn how
62 to find datasets, load them efficiently, interpret metadata, operate on multi-dimensional
63 arrays, and produce scientifically meaningful diagnostics. General-purpose libraries such
64 as xarray (Hoyer & Hamman, 2017) provide the foundations for these tasks, but they
65 do not by themselves show a learner how to translate a research question into a robust
66 analysis workflow for a specific model and computing environment.

67 The COSIMA Cookbook fills that need with a domain-specific, openly maintained collection
68 of computational lessons and examples. Its contribution is not a new analysis package;
69 rather, it is a reusable learning resource that makes existing scientific software, data
70 conventions, and model output more approachable. This aligns well with JOSE's emphasis
71 on open educational resources that enable computational learning through authentic
72 practice.

73 Several design decisions make the Cookbook particularly useful for adoption by other groups.
74 First, each notebook is intended to be self-contained and well-documented, so learners
75 can read, run, and modify a complete workflow without reconstructing missing context
76 from scattered notes. Second, the repository distinguishes tutorials from recipes. Tutorials
77 teach transferable skills, while recipes demonstrate how those skills combine in realistic
78 analysis tasks. Third, the project includes contributor guidance and notebook review
79 conventions that encourage new analyses to be written in a reusable and pedagogically
80 clear style.

81 This structure is valuable beyond the immediate COSIMA context. Many research
82 communities maintain model output on shared infrastructure and face the same challenge:
83 turning expert tacit knowledge into examples that newcomers can adapt. The Cookbook
84 offers a model for how a scientific collaboration can capture that knowledge in version-
85 controlled notebooks, publish it as a living resource, and continuously improve it through
86 community contribution.

87 Educational Design and Experience of Use

88 The learning experience is organised as a progression. The documentation points new users
89 first to introductory lessons, including material on loading, slicing, and visualising model
90 output. Learners then move to more advanced tutorials and finally to recipe notebooks
91 that address concrete scientific questions. The categories of “appetisers”, “mains”, and
92 “local dishes” provide a lightweight pedagogical cue about expected complexity and scope,
93 with “local dishes” specifically covering recipes for regional model configurations (Barnes
94 et al., 2024).

95 The intended mode of use is also explicit. The repository is designed around Jupyter-
96 based analysis on the Australian National Computational Infrastructure, with guidance
97 on running notebooks through ARE/JupyterLab sessions and on accessing shared data
98 holdings. This makes the Cookbook more than a static collection of examples: it is
99 operational documentation for a real analysis environment. At the same time, the

notebooks demonstrate broadly transferable patterns for working with labelled geophysical data, so individual lessons can be reused or adapted outside that environment.

The project also supports community authorship. Contributors are encouraged to submit new notebooks through pull requests, document their methods clearly, and generalise workflows so that they remain useful to others. In that sense, the Cookbook functions both as a learning module for end users and as a framework for teaching reproducible scientific communication through notebook design and peer review.

Author order

The first two authors contributed equally; authors from the third position onward are listed alphabetically by surname.

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References

- Barnes, A. J., Constantinou, N. C., Gibson, A. H., Kiss, A. E., Chapman, C., Reilly, J., Bhagtani, D., & Yang, L. (2024). Regional-mom6: A python package for automatic generation of regional configurations for the modular ocean model 6. *Journal of Open Source Software*, 9(100), 6857. <https://doi.org/10.21105/joss.06857>
- COSIMA contributors. (2026). *COSIMA cookbook: A cookbook of recipes for analysing ocean and sea ice model output*. <https://doi.org/10.5281/zenodo.14353852>
- Hoyer, S., & Hamman, J. J. (2017). Xarray: N-D labeled arrays and datasets in Python. *Journal of Open Research Software*, 5(1), 10. <https://doi.org/10.5334/jors.148>
- Intake developers. (2026). *Intake: A lightweight package for finding, investigating, loading and disseminating data*. <https://intake.readthedocs.io/en/reader/>
- Kiss, A. E., Woodward, M. J., Hollings, T., Mu, M., Fiedler, R., Steven, A. D. L., Lenton, A., Matear, R., Chamberlain, M. A., Oke, P. R., Alves, O., Griffies, S. M., Hill, C., Kovach, A., Marsland, S. J., Fiedler, R., Waters, J., Yang, C.-H., Zhou, X., ... Hogg, A. McC. (2020). ACCESS-OM2 v1.0: A global ocean-sea ice model at three resolutions. *Geoscientific Model Development*, 13(2), 401–442. <https://doi.org/10.5194/gmd-13-401-2020>
- Kluyver, T., Ragan-Kelley, B., Pérez, F., Granger, B., Bussonnier, M., Frederic, J., Kelley, K., Hamrick, J., Grout, J., Corlay, S., Ivanov, P., Avila, D., Abdalla, S., & Willing, C. (2016). Jupyter notebooks—a publishing format for reproducible computational workflows. In F. Loizides & B. Schmidt (Eds.), *Positioning and power in academic publishing: Players, agents and agendas* (pp. 87–90). IOS Press. <https://doi.org/10.3233/978-1-61499-649-1-87>
- Lange, M., & Seville, E. van. (2017). Parcels v0.9: Prototyping a Lagrangian ocean

- 143 analysis framework for the petascale age. *Geoscientific Model Development*, 10(11),
144 4175–4186. <https://doi.org/10.5194/gmd-10-4175-2017>
- 145 Met Office. (2010--2015). *Cartopy: A cartographic Python library with a Matplotlib*
146 *interface*. <https://cartopy.readthedocs.io>
- 147 xgcm Developers. (2025). *Xgcm: General circulation model postprocessing with xarray*.
148 <https://xgcm.readthedocs.io/en/latest/>

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